Benchmarking Open-Weight Foundation Models for Global AI Technical Governance

Jason Hung
Internet Society

Abstract**—**Large language models (LLMs) are increasingly deployed in artificial intelligence (AI) governance analysis across national and international organisations. Existing studies face three methodological limitations: (1) reliance on proprietary models, preventing independent replication; (2) evaluation of years beyond training cutoffs, measuring temporal ignorance rather than geographic bias; and (3) binary classification that conflates confident fabrication (HF) with honest acknowledgement of uncertainty. This study addresses all three by benchmarking four open-weight frontier models against the Global AI Dataset v2 (GAID v2), a verified ground-truth database of 24,453 indicators across 227 countries published on Harvard Dataverse in January 2026. A total of 18 indicators mapped to the eight IEEE IRAI 2026 thematic dimensions yield approximately 2,990 country-metric-year observations (2010–2023). A five-category classifier distinguishes verified accuracy (VF), HF, honest refusal (HR), qualitative hedging (QH), and misattribution (MF). Geographic disparities are estimated via mixed-effects logistic regression and difference-in-differences (DiD) analysis. Contrary to the dominant hypothesis, fabrication rates are higher for Global North than Global South countries, which we speculate is driven by indicator scale and specificity but not training-data underrepresentation.

Index Terms**—**geographic bias, large language models, AI governance, open-weight models, hallucination, factual accuracy, Global South, IEEE IRAI.

I. INTRODUCTION

A. Overview and Research Problem

Large language models (LLMs) are now routinely used to synthesise cross-national artificial intelligence (AI) policy information [4]. Their reliability depends on factual accuracy across all countries, yet existing geographic bias studies suffer from three limitations: (1) use of proprietary models whose outputs change without notice, barring replication [11]; (2) evaluation of years after model training cutoffs, measuring temporal ignorance rather than geographic bias [3]; and (3) binary correct-or-incorrect classification that conflates confident fabrication (HF) with honest acknowledgement of uncertainty [6]. This study addresses all three through a fully replicable, open-weight evaluation against a verified ground-truth benchmark.

B. Research Questions

RQ1. Do the four models exhibit statistically significant geographic variation in accuracy?

RQ2. Does the developer's national origin moderate geographic accuracy differentials?

RQ3. Do differentials vary systematically across the eight IEEE IRAI 2026 thematic dimensions [5]?

RQ4. Does proximity to training cutoff independently account for observed differentials, once geographic and thematic factors are controlled?

II. LITERATURE REVIEW

A. Geographic and Cultural Bias in LLMs

Substantial empirical literature documents that LLMs encode the distributional characteristics of training corpora, systematically favouring wealthy, English-speaking, and geopolitically significant countries [10], [11]. Arora et al. [1] show pre-trained models exhibit weaker alignment with non-Western values; Cao et al. [2] demonstrate geopolitical bias in analytical framing; Navigli et al. [15] identify a lingua franca bias linking factual accuracy to the linguistic prominence of the queried country. Ji et al. [6] identify geographic sparsity as a principal driver of extrinsic hallucination for factual queries. This study examines geographic bias directly within the AI governance domain [4].

B. Methodological Limitations in Prior Work

Most high-profile evaluations rely on proprietary, closed-source systems [11] preventing independent replication [7]. Temporal limitations mean many studies conflate training-cutoff ignorance with geographic bias [3]. Most studies classify responses into a binary correct-or-incorrect scheme [11], failing to distinguish HF from honest uncertainty—a distinction with substantial implications for how geographic bias should be understood. This study exclusively employs open-weight models with documented version identifiers, evaluates within training periods, and applies a five-category classifier.

III. METHODOLOGY

A. Overview

The evaluation proceeds through five stages: (1) 2,990 country-indicator-year observations are extracted from Global AI Dataset v2 (GAID v2) across 18 indicators and six evaluation years; (2) three query variants are generated per observation; (3) prompts are submitted to all four models at temperature zero, yielding 35,880 responses; (4) each response is scored using the five-category classifier; and (5) classified responses are passed to three statistical procedures—mixed-effects logistic regression, difference-in-differences (DiD) estimation, and principal component analysis (PCA). Results are disseminated via

this conference paper and an open-source methodology release.

B. Ground-Truth Data: GAID v2

The GAID v2, published on Harvard Dataverse in January 2026, contains 259,546 rows covering 24,453 indicators across 227 countries (1998–2025), drawing on 11 curated international sources including the Stanford AI Index, WIPO, Epoch AI, World Bank GovTech Maturity Index, Coursera, and Ipsos. Each row records the verified value for a given indicator in a given country and year, treated as ground truth for evaluation.

C. Model Selection

Four open-weight frontier models are evaluated (Table 1), achieving geopolitical balance between two Western-developer models (Llama 4 Maverick, Mistral Large 3) and two Chinese-developer models (Qwen3-235B-A22B, DeepSeek-V3-0324). All are available on Hugging Face with fixed parameter weights, providing reproducibility guarantees that proprietary API-based evaluations cannot. DeepSeek-R1 is excluded because its chain-of-thought response structure is architecturally incompatible with the structured factual query format used in this study.

TABLE 1

Four open-weight frontier language models evaluated in this study

Model	Developer	Architecture	Training End
Llama 4 Maverick	Meta AI (US)	MoE — 17B active / 400B total	August 2024 [12] [13]
Mistral Large 3	Mistral AI (France)	MoE — ~41B active / 675B total	By late 2025 (est.) [14]
Qwen3-235B -A22B	Alibaba Cloud (China)	MoE — 22B active / 235B total	By mid 2025 (est.) [16]
DeepSeek-V 3-0324	DeepSeek (China)	MoE — 37B active / 671B total	July 2024 [8]

D. Evaluation Years and Query Design

Six evaluation years are used: 2010, 2013, 2016, 2019, 2022, and 2023. Years 2010–2019 are designated the primary analysis window, falling within all four models' training periods; 2022–2023 serve as a robustness check for training-proximity effects. For each of the approximately 2,990 individual observations, three query variants are generated: a direct numeric query (Variant 1), a comparative framing embedding the target country in a regional context (Variant 2), and a contextual governance narrative framing (Variant 3).

E. Response Classification

Each response is classified into five mutually exclusive categories using a fully automated, rule-based classifier (Table 2). The proportion classified as HF constitutes the primary outcome variable—the fabrication rate. A random five percent sample was manually reviewed to verify classifier accuracy (aggregate consistency rate: 90.1 percent across all four models).

TABLE 2

Five-category response classification scheme with definitions

Code	Category	Definition	Significance
VF	Verified Accurate	Numeric estimate within ±10% of verified GAID v2 value.	Baseline for geographic accuracy comparisons.
HF	Confident Fabrication	Numeric estimate outside ±10% threshold; factually incorrect.	Primary indicator of geographic bias.
HR	Honest Refusal	Explicitly acknowledges absence of specific knowledge.	Indicates calibrated uncertainty.
QH	Qualitative Hedging	Directional/comparative description without numeric commitment.	Secondary bias indicator.
MF	Misattributi on	Responds with information about a different country/year.	Indicates systematic deflection.

F. Statistical Analysis

The fabrication rate is modelled as a binary outcome via logistic regression with random intercepts for country, nested within world region. Fixed-effect predictors include world region, model identity, evaluation year, and IEEE IRAI thematic dimension:

$$logit(P[HF_{ijkmt}]) = \beta_0 + \beta_1 Region_j + \beta_2 Model_k + \beta_3 Year_t + \beta_4 Theme_m + u_j + \varepsilon_{ijkmt}$$

The geopolitically balanced model selection enables a DiD design estimating whether the Global South accuracy advantage differs across developer origin groups. PCA is applied to the 177-country × 18-indicator VF rate matrix to assess the dimensionality of geographic bias.

IV. DATASET AND EXPERIMENT PLAN

A. Indicators Selected from GAID v2

The 18 indicators (Table 3) were selected through a three-stage screening process: (i) thematic mapping to the eight IEEE IRAI 2026 dimensions; (ii) coverage analysis retaining indicators with verified values for at least ten countries across at least one evaluation year; and (iii) redundancy screening excluding indicators with Spearman r > 0.90 with another selected indicator. Together they yield approximately 2,990 country-metric-year observations.

TABLE 3

18 indicators selected from GAID v2, mapped to IEEE IRAI 2026 thematic dimensions

IRAI Theme	Code	Indicator Name	Source
Transparency	*AI_Pubs*	Number of AI Publications: All	Stanford AI Index / Scopus
Transparency	*WIPO_Patents*	Total AI-Related Patent Publications	WIPO
Transparency	*FWCI*	Field Weighted Citation Impact: All	Stanford AI Index / Scopus
Fairness	*CS_Grad_F*	Share of CS Bachelor's Graduates Who Are Female	Stanford AI Index / OECD
Fairness	*CS_PhD_F*	Share of CS Doctoral Graduates Who Are Female	Stanford AI Index / OECD
Adoption	*WB_GTMI*	World Bank GovTech Overall Maturity Index	World Bank GTMI 2022

IRAI Theme	Code	Indicator Name	Source
Adoption	*Coursera_Biz*	Coursera Business Skills Proficiency Score	Coursera Global Skills Report
Adoption	*Coursera_Tech*	Coursera Technology Skills Proficiency Score	Coursera Global Skills Report
Adoption	*BigData_Biz*	Proportion of Businesses Performing Big Data Analysis	OECD.ai
Regulation	*Nat_AI_Strat*	National AI Strategy Published (binary)	Stanford AI Index
Regulation	*AI_Bills*	Number of AI-Related Bills Enacted into Law	Stanford AI Index
Accountability	*AI_Legis*	AI References in National Legislative Proceedings	Stanford AI Index
Safety	*Train_Compute*	Total Training Compute for AI Models Released (FLOP)	Epoch AI
Safety	*Model_Params*	Total Parameters of AI Models Released	Epoch AI
Security	*ICT_Sec_All*	Proportion of Businesses That Experienced ICT Security Breaches	OECD.ai
Security	*ICT_Sec_ICT*	ICT Sector Businesses That Experienced ICT Security Breaches	OECD.ai
Ethics	*AI_Benefits*	Survey Respondents: AI Offers More Benefits Than Drawbacks	Stanford AI Index / Ipsos
Ethics	*AI_Nervous*	Survey Respondents: AI Products Make Them Nervous	Stanford AI Index / Ipsos

B. Dataset Limitations

Six limitations of GAID v2 are acknowledged. Ethics indicators (*AI_Benefits, AI_Nervous*) derive exclusively from the 2023 Ipsos survey, precluding longitudinal analysis. Security indicators cover approximately 36 OECD member states, excluding virtually all of the Global South. Fairness indicators cover only 16–21 countries. Accountability relies on a single indicator; the closest alternative candidate correlates at r = 0.906 with *AI_Legis*, exceeding the redundancy threshold. *WB_GTMI* is available only for 2022. Mistral Large 3 and Qwen3-235B-A22B do not have publicly disclosed precise training data end dates; findings for these models are interpreted within available data constraints.

V. RESULTS

A. Overall Classification Distribution

Across 35,880 queries (2,990 observations × 3 variants × 4 models), 27.4 percent of responses were classified as VF and 71.8 percent as HF. Honest refusal (HR) was effectively absent (<0.1 percent), indicating frontier LLMs almost never decline to answer factual queries regardless of whether they possess the relevant knowledge. Query framing significantly affected fabrication rates: comparative framing (Variant 2) reduced HF odds by 41 percent (OR=0.59, p<0.001) and contextual framing (Variant 3) by 18 percent (OR=0.82, p<0.001) relative to direct numeric queries.

B. Model-Level Variation

Fabrication rates varied substantially across models (Fig. 1). Mistral Large 3 recorded the highest verified accuracy (VF) rates (Global North: 29.7%; Global South: 44.5%), followed by DeepSeek-V3-0324 (25.1%; 37.2%), Qwen3-235B-A22B (19.6%; 30.6%), and Llama 4 Maverick (11.0%; 15.9%). Model identity is the strongest single predictor: relative to DeepSeek-V3-0324, Llama 4 Maverick shows 2.40 times the odds of fabrication (OR=2.40, p<0.001), Qwen3 1.49 times (OR=1.49, p<0.001), and Mistral 0.78 times (OR=0.78, p<0.001). The rank ordering is stable across primary and secondary analysis windows.

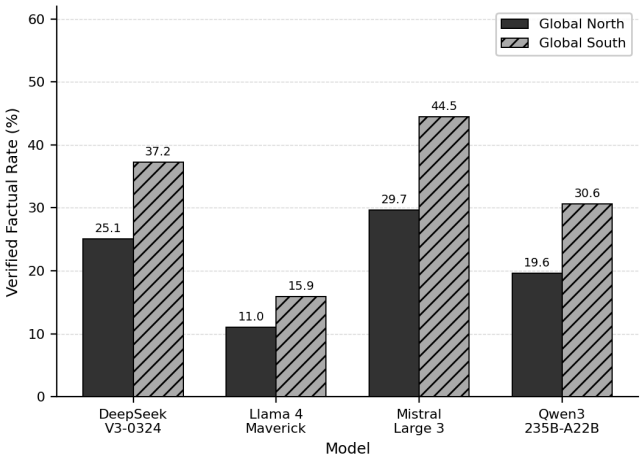

Fig. 1. VF Rate by Model and Global North/South (2010–2023)

C. Regional Variation (RQ1)

Fabrication rates varied across regions in a direction contrary to the primary hypothesis (RQ1 confirmed; pattern inverted). Africa recorded the highest mean VF rate (35.1%), followed by the Americas (30.2%), Asia (28.1%), Europe (24.4%), and Oceania (23.5%). In the mixed-effects regression, Africa showed significantly lower fabrication odds than Europe, the reference category (OR=0.52, 95% CI [0.38, 0.71], p<0.001), as did the Americas (OR=0.66, p=0.012). Asia approached significance (OR=0.79, p=0.063). Oceania did not differ significantly from Europe (OR=0.94, p=0.850).

D. Developer Origin and Geographic Disparity (RQ2)

Fig. 2 presents the 2×2 HF rates disaggregated by developer origin. Western-origin models produce HF rates of 68.8% for Global South versus 78.8% for Global North (−10.0 pp); Chinese-origin models produce 65.2% versus 77.2% (−12.0 pp). The DiD estimator yields +2.07 percentage points (OR=1.090, SE=0.049, z=1.776, p=0.076): marginal evidence that developer origin modestly moderates the Global South accuracy advantage (RQ2: marginal). The developer-origin main effect is negligible (OR=1.003, p=0.941).

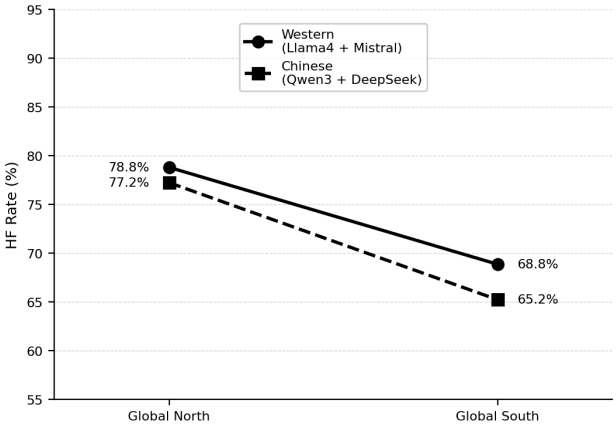

Fig. 2. DiD: Developer Origin × Country Income (2010–2023). Raw DiD = +2.07 pp

E. Thematic Variation (RQ3)

Fabrication rates differed substantially across IEEE IRAI themes (Fig. 3). The Safety theme—comprising *Train_Compute* and *Model_Params*—recorded a mean VF rate of 1.8 percent; the Safety effect is the largest in the regression (OR=11.93 relative to Transparency, 95% CI [6.16, 23.11], $p<0.001$), showing near-total fabrication of AI compute parameters. The Regulation theme recorded the highest mean VF rate (42.2%), with significantly lower fabrication odds relative to Transparency (OR=0.34, $p<0.001$). Thematic rankings are consistent across all four models.

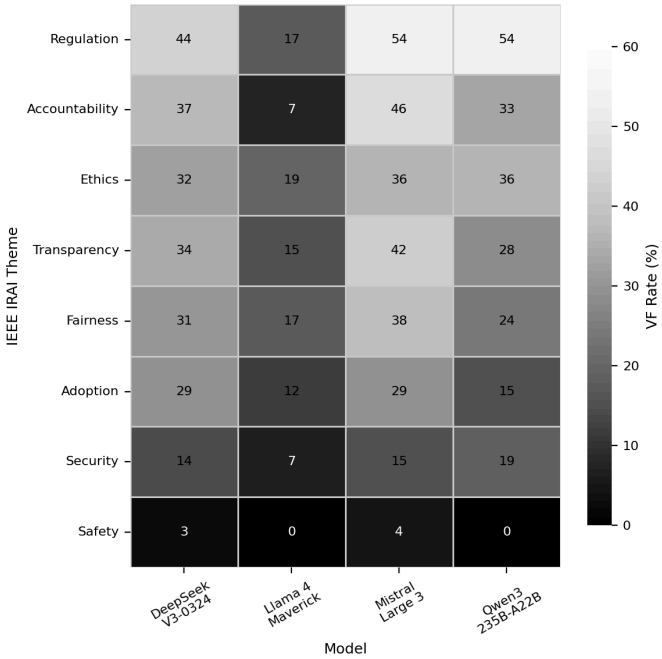

Fig. 3. VF Rate (%) by IEEE IRAI Theme and Model (2010–2023)

F. Principal Component Structure of Geographic Bias (RQ3)

PCA applied to the 177-country × 18-indicator VF rate matrix (Fig. 4) requires 15 components to explain 90 percent of variance (PC1: 11.3%; cumulative through PC15: 92.9%). PC1 contrasts Regulation and Safety indicators against Ethics and Adoption (*AI_Bills*: +0.373; *Train_Compute*: +0.348; *AI_Nervous*: −0.370; *WB_GTMI*: −0.328). PC2 shows mixed Accountability, Security, Transparency, and Ethics loadings (*AI_Legis*: +0.379; explains 10.4%). No single component accounts for more than 11.3% of variance; the biplot (Fig. 5) shows no cluster separating Global North from Global South. Geographic accuracy differentials are thematically structured but diffuse.

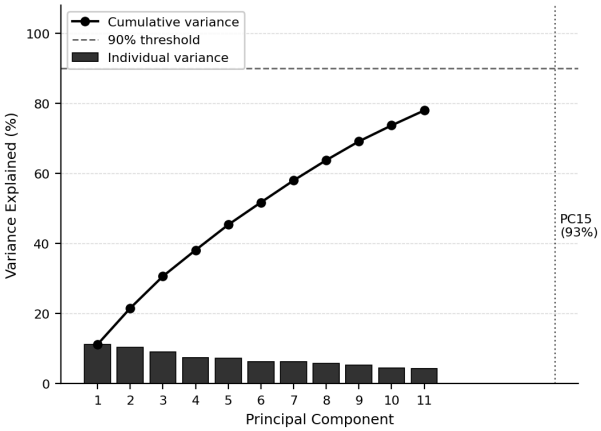

Fig. 4. PCA Scree Plot — Country × Indicator VF Matrix (2010–2023)

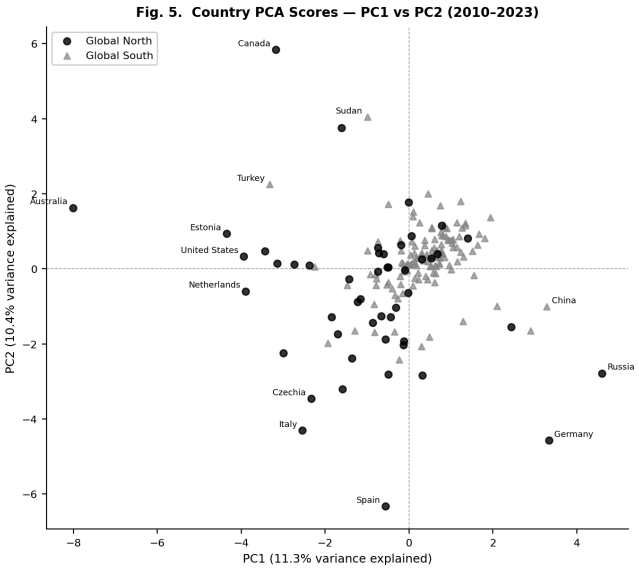

Fig. 5. Country PCA Scores — PC1 vs PC2 (2010–2023)

G. Temporal Effects (RQ4)

Within the primary analysis window (2010–2019), VF rates were highest in 2010 (30.7%) and lowest in 2016 (25.3%), meaning fabrication peaked in 2016 and followed a non-monotonic trend (OR=1.30 for 2016 and OR=1.21 for 2019 relative to the 2010 baseline (p=0.009; p=0.068, respectively), confirming recovery from the 2016 peak). In the secondary window (2022–2023), VF rates of 24.2% and 27.6% are consistent with 2016–2019, confirming that training-period proximity does not independently account for observed patterns (RQ4: not confirmed).

VI. DISCUSSION

A. The Inverted Geographic Pattern

The most surprising finding is that fabrication rates are higher for Global North countries, contradicting the dominant hypothesis that Global South underrepresentation drives fabrication [9], [11]. Two structural features account for this. First, the ±10% proportional accuracy threshold structurally favours small-value metrics disproportionately found in Global South countries. A model estimating 40 AI publications when the verified value is 44 is classified as VF; the same model responding that the United States produced 12,000 publications, when the verified value is 15,000, is classified as HF. Second, the GAID v2 observation frame is not geographically neutral—AI compute and patent data are populated almost exclusively by a small number of high-income countries, making these the highest-specificity queries for which fabrication rates are highest. These findings do not imply geographic bias is absent; they suggest its empirical manifestation is more complex than a simple Global North advantage.

B. Thematic Specificity and PCA Structure

Near-total fabrication for Safety indicators suggests a structural limitation of current LLMs: AI compute and parameter data exist at a scale and specificity outside model knowledge bases entirely. The PCA results confirm that geographic bias does not reduce to a single underlying dimension; 15 components explain 90 percent of variance, and dominant contrasts are thematic (Regulation and Safety in PC1, Accountability and Security in PC2) rather than a simple Global North/South gradient (Fig. 5). A country's accuracy profile depends more on which indicators are queried and what the verified values are than on a general geographic factor.

C. Developer Origin and Query Framing

The marginal DiD estimate (+2.07 pp, p=0.076) provides limited evidence that developer origin modestly moderates geographic accuracy differentials; the effect is small and does not survive correction for multiple comparisons. Comparative framing reduces fabrication odds by 41 percent, suggesting query structure is a more powerful lever for practitioners than model choice. No currently available open-weight frontier model can be relied upon for precise quantitative AI governance data; Mistral Large 3, the most accurate model examined, fabricated 61.2 percent of responses. Human verification against primary sources—Stanford AI Index, WIPO, World Bank GovTech—remains essential.

D. Limitations and Future Directions

Four limitations should be acknowledged. First, the ±10% proportional threshold confounds scale with accuracy; future studies should consider scale-normalised thresholds or separate analyses by indicator value distribution. Second, quantised (*FP8*) API deployments for two models may affect factual precision at the margin. Third, the Global North/South binary conceals substantial within-group heterogeneity; finer country-income sub-categories would provide a more differentiated account. Fourth, HR responses are effectively absent across all frontier models (<0.1%), leaving users no behavioural signal to tell HF from high-confidence output. Such a pattern differs noticeably from smaller models and creates a practical safety gap for AI governance practitioners.

REFERENCES

[1] A. Arora, L. Kaffee, and I. Augenstein, "Probing pre-trained language models for cross-cultural differences in values," in Proc. C3NLP at EACL 2023, pp. 12–25. Available: https://aclanthology.org/2023.c3nlp-1.12

[2] Y. Cao et al., "Assessing cross-cultural alignment between ChatGPT and human societies," in Proc. C3NLP at EACL 2023, pp. 53–67. Available: https://aclanthology.org/2023.c3nlp-1.7

[3] J. Cheng et al., "Dated data: Tracing knowledge cutoffs in large language models," arXiv, 2024. doi: 10.48550/arXiv.2403.12958

[4] J. Hung, "Global AI bias audit for technical governance," arXiv, 2026. doi: 10.48550/arXiv.2602.13246

[5] IEEE Industrial Electronics Society, "IEEE IRAI 2026: Call for papers," 2026. Available: https://irai2026.ieee-ies.org/

[6] Z. Ji et al., "Survey of hallucination in natural language generation," ACM Comput. Surv., vol. 55, no. 12, Art. 248, 2023. doi: 10.1145/3571730

[7] S. Kapoor et al., "REFORMS: Consensus-based recommendations for machine learning-based science," Sci. Adv., vol. 10, no. 18, Art. eadk3452, 2024.

[8] Knostic, "DeepSeek's cutoff date is July 2024," Feb. 2025. Available: https://www.knostic.ai/blog/exposing-deepseek-system-prompts

[9] H. Lalai et al., "The world according to LLMs: How geographic origin influences LLMs' entity deduction capabilities," arXiv, 2025. doi: 10.48550/arXiv.2508.05525

[10] Z. Li, "Investigating spatial-temporal bias of LLMs," Expert Syst. Appl., vol. 313, Art. 131542, 2026.

[11] R. Manvi et al., “Large language models are geographically biased,” arXiv, 2024. doi: 10.48550/arXiv.2402.02680

[12] Meta, “Llama-4-Maverick-17B-128E-Original,” Hugging Face, 2025. Available: https://huggingface.co/meta-llama/Llama-4-Maverick-17B-128E-Original

[13] Meta, “The Llama 4 herd,” Meta AI Blog, Apr. 2025. Available: https://ai.meta.com/blog/llama-4-multimodal-intelligence/

[14] Mistral AI, “Mistral Large 3 (675B Instruct),” NVIDIA Build API Catalog, 2025. Available: https://build.nvidia.com/mistralai/mistral-large-3-675b-instruct-2512/modelcard

[15] R. Navigli, S. Conia, and B. Ross, “Biases in large language models: Origins, inventory, and discussion,” ACM J. Data Inf. Qual., vol. 15, no. 2, pp. 1–21, 2023.

[16] Qwen, “Qwen3: Think deeper, act faster,” Qwen Blog, Apr. 2025. Available: https://qwenlm.github.io/blog/qwen3/